Survey Methods
1.1 – The scope of survey research

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Stellenbosch
UNIVERSITY
IYUNIVESITHI
UNIVERSITEIT
forward together
sonke siya phambili
saam vorentoe

Outline

- The nature of surveys
- Quantitative vs qualitative
- Surveys, theories and paradigms

What is a survey?

https://tinyurl.com/smethods

05:00

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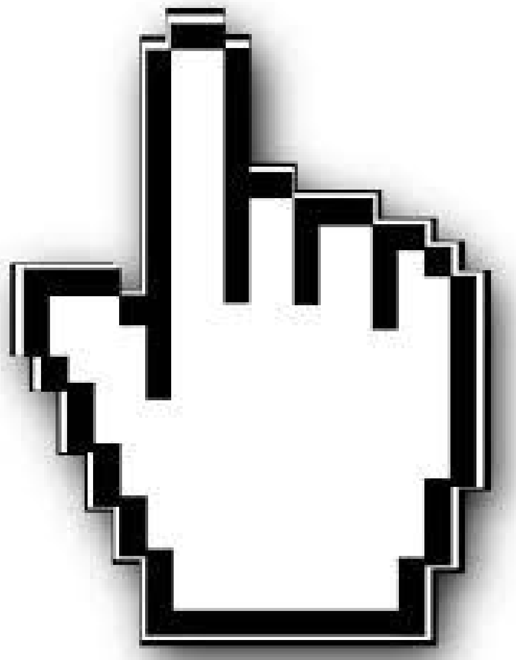
Online Results



A formal definition

A "survey" is a systematic method for gathering information from (a sample of) entities for the purposes of constructing quantitative descriptors of the attributes of the larger population of which the entities are members.

(Groves et al. 2009:3)

Planned, documented, formalised

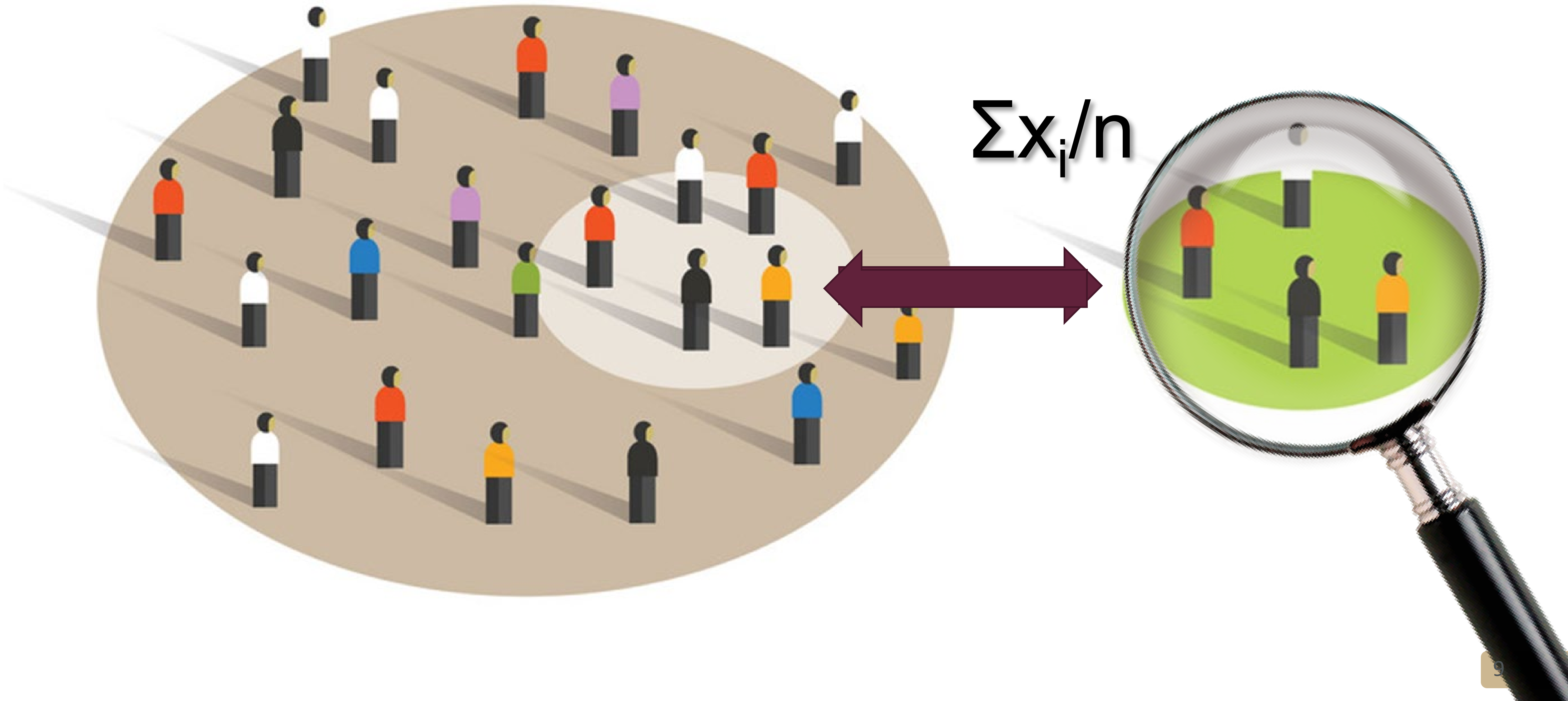
People, facilities, countries…

A "survey" is a systematic method for gathering information from (a sample of) entities for the purposes of constructing quantitative descriptors of the attributes of the larger population of which the entities are members.

Not **one** entity

Statistics (descriptive, analytic)

A sample...



What characterises a survey?

Form of data

Surveys are characterised by a structured or systematic set of data that can be arranged in a grid

Logic of analysis

Data are analysed by comparing observed characteristics of cases

PROV	SEX	POPGROUP	AGE	AGECAT10	HEIGHT	WEIGHT	WAIST	BMI
GT	Female	White	66	(65,75]	1.63	72.7	104.5	27.5
GT	Male	White	26	(25,35]	1.61	55.8	64.2	21.5
GT	Female	White	84	(75,120]	1.56	75.2	108.0	30.9
GT	Female	White	46	(45,55]	1.70	108.0	110.0	37.4
GT	Female	White	52	(45,55]	1.61	107.0	109.0	41.3
MP	Male	White	17	(14,25]	1.76	72.5	78.0	23.4
GT	Male	White	68	(65,75]	1.60	49.3	58.2	19.3
GT	Female	White	25	(14,25]	1.62	66.5	78.1	25.3
GT	Female	White	41	(35,45]	1.63	74.2	100.5	28.1
GT	Male	White	62	(55,65]	1.71	87.1	108.5	29.8
GT	Female	White	63	(55,65]	1.56	69.9	99.2	28.7
GT	Female	White	51	(45,55]	1.59	105.0	111.0	41.5
GT	Male	White	38	(35,45]	1.68	83.6	100.0	29.6
GT	Female	White	69	(65,75]	1.67	106.0	119.0	38.0
GT	Male	White	16	(14,25]	1.62	49.3	65.3	18.8
GT	Male	White	34	(25,35]	1.87	90.2	101.5	25.8
GT	Female	Asian	37	(35,45]	1.60	64.3	74.0	25.1
GT	Male	Asian	35	(25,35]	1.71	59.0	77.2	20.1
GT	Male	Asian	17	(14,25]	1.69	47.6	58.7	16.6
GT	Male	White	82	(75,120]	1.65	56.3	97.3	20.8
GT	Female	White	20	(14,25]	1.70	90.5	89.2	31.3

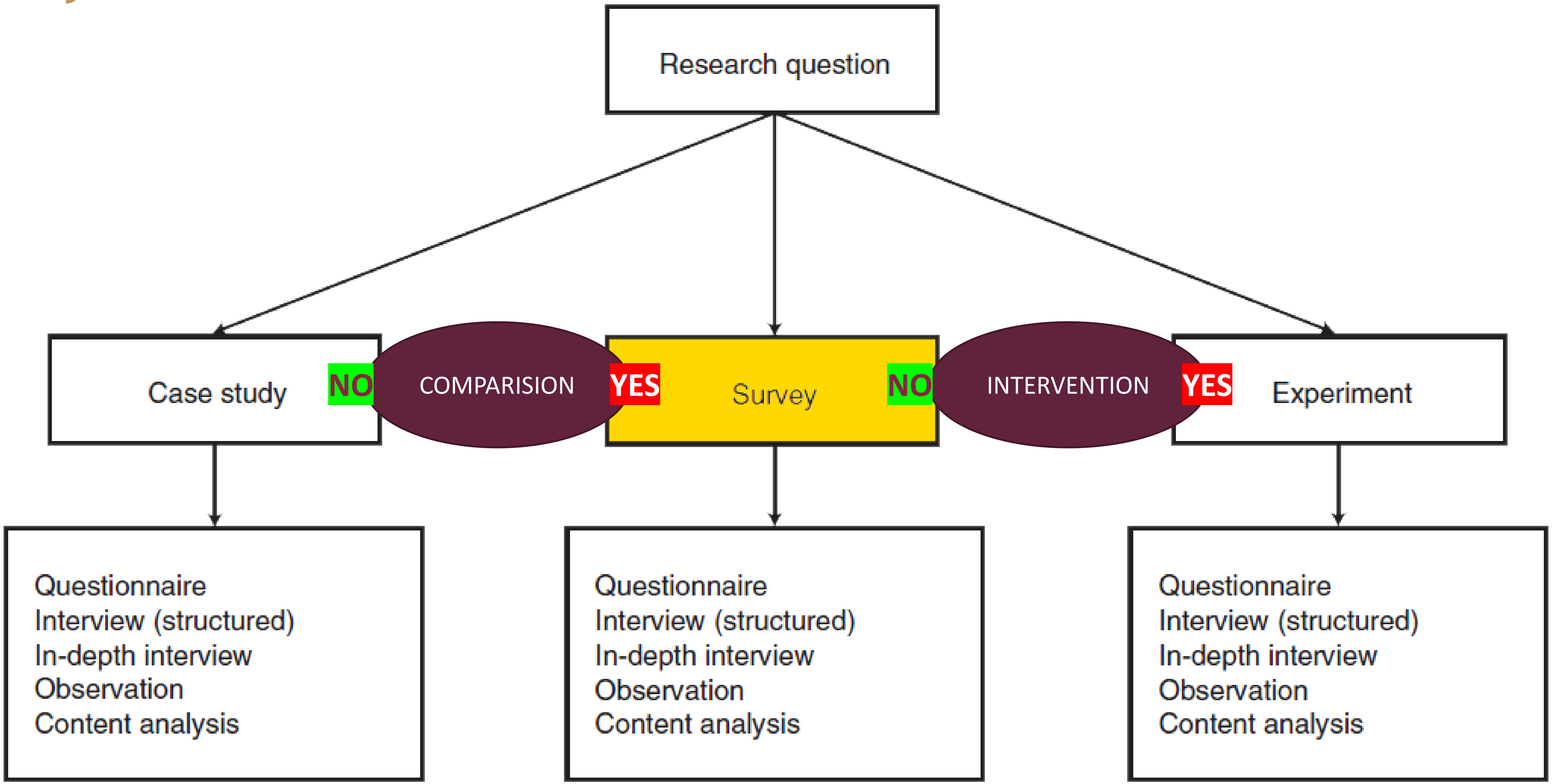
Form of data

Variables

Cases

PROV	SEX	POPGROUP	AGE	AGECAT10	HEIGHT	WEIGHT	WAIST	BMI	BMICAT	CESD10	CLUSTER	STRATUM	PWEIGHT
GT	Female	White	66	(65,75]	1.63	72.7	104.5	27.5	Overweight	17	7001	88	8911.878
GT	Male	White	26	(25,35]	1.61	55.8	64.2	21.5	Normal Weight	14	7001	88	8911.878
GT	Female	White	84	(75,120]	1.56	75.2	108.0	30.9	Obese	24	7001	88	8911.878
GT	Female	White	46	(45,55]	1.70	108.0	110.0	37.4	Obese	19	7001	88	1103.276
GT	Female	White	52	(45,55]	1.61	107.0	109.0	41.3	Obese	10	7001	88	2318.409
MP	Male	White	17	(14,25]	1.76	72.5	78.0	23.4	Normal Weight	20	8015	31	1660.189
GT	Male	White	68	(65,75]	1.60	49.3	58.2	19.3	Normal Weight	13	7001	88	3027.722
GT	Female	White	25	(14,25]	1.62	66.5	78.1	25.3	Overweight	30	7001	88	2487.846
GT	Female	White	41	(35,45]	1.63	74.2	100.5	28.1	Overweight	22	7001	88	2235.242
GT	Male	White	62	(55,65]	1.71	87.1	108.5	29.8	Overweight	19	7001	88	2236.405
GT	Female	White	63	(55,65]	1.56	69.9	99.2	28.7	Overweight	14	7001	88	2236.405
GT	Female	White	51	(45,55]	1.59	105.0	111.0	41.5	Obese	21	7001	88	1543.711
GT	Male	White	38	(35,45]	1.68	83.6	100.0	29.6	Overweight	19	7001	88	1103.276
GT	Female	White	69	(65,75]	1.67	106.0	119.0	38.0	Obese	19	7001	88	3027.722
GT	Male	White	16	(14,25]	1.62	49.3	65.3	18.8	Normal Weight	15	7021	773	4544.761
GT	Male	White	34	(25,35]	1.87	90.2	101.5	25.8	Overweight	14	7021	773	4533.682
GT	Female	Asian	37	(35,45]	1.60	64.3	74.0	25.1	Overweight	30	7021	773	22306.74
GT	Male	Asian	35	(25,35]	1.71	59.0	77.2	20.1	Normal Weight	28	7021	773	3363.536
GT	Male	Asian	17	(14,25]	1.69	47.6	58.7	16.6	Underweight	13	7021	773	3363.536
GT	Male	White	82	(75,120]	1.65	56.3	97.3	20.8	Normal Weight	13	7021	773	2434.573
GT	Female	White	20	(14,25]	1.70	90.5	89.2	31.3	Obese	10	7021	773	4533.682

Analysis



(Figure adapted from: De Vaus, 2002:6)

Origins and development

1930 – 1960 Invention	1960 – 1990 Expansion	1990 + Designed + Organic data
• Sampling frames with universal coverage • Probability sampling • Structured questions • Statistical inference to finite populations with measurable sampling error	• Telephonic sampling frames • Computers, CATI, CAPI • University training • Cognitive theories • Declining response rates	• Decline in traditional sampling frames • New technologies: mobile phones, Internet • Emergence of “organic data” • Big data

(Groves 2011)

Polls & Surveys

What is the difference between a poll and a survey? The word "poll" is most often used for private-sector opinion studies, which use many of the same design features as studies that would be called "surveys." "Poll" is rarely used to describe studies conducted in government or scientific domains. There are, however, no clear distinctions between the meanings of the two terms. Schuman notes that the two terms have different roots: "'Poll' is a four letter word, generally thought to be from an ancient Germanic term referring to 'head,' as in counting heads. The two-syllable word 'survey,' on the other hand, comes from the French *survee*, which in turn derives from Latin *super* (over) and *videre* (to look). The first is therefore an expression with appeal to a wider public, the intended consumers of results from Gallup, Harris, and other polls. The second fits the needs of academicians in university institutes who wish to emphasize the scientific or scholarly character of their work."

(Groves et al. 2009:4. Citing Shuman 1997)

Imagine that you believe a diet too high on sugars increases the risk of diabetes.

How would you test this theory by using:

1) a case study
2) an experiment
3) a survey

10:00

Start Stop Reset mins: 10 secs: 0 type: None

Survey Methods

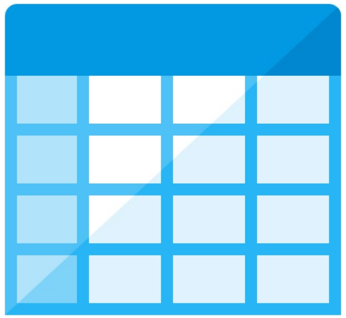
Does our ‘mini’ data collection falls within the definition of survey?

A "survey" is a systematic method for gathering information from (a sample of) entities for the purposes of constructing quantitative descriptors of the attributes of the larger population of which the entities are members.

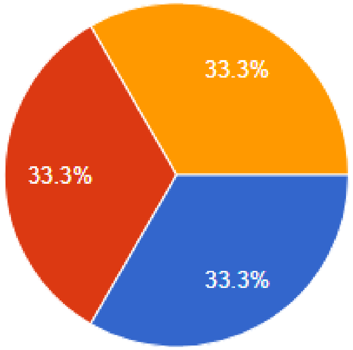
Data Dictionary Codebook

#	Variable / Field Name	Field Label *Field Note*	Field Attributes (Field Type, Validation, Choices, Calculations, etc.)
Instrument: **Basic Demography Form** (demographics) Enabled as survey			
1	[record_id]	Study ID	text
2	[sex]	Sex	radio, Required 1 Female 2 Male 3 Prefer not to say
3	[age]	Age (years)	text (integer, Min: 18, Max: 99), Required
4	[background]	Background	radio, Required 1 Social sciences 2 Medicine 3 Psychology 4 Epidemiology 5 Statistics 6 Other
5	[wellbeing]	How do you feel today?	slider (Min: 0, Max: 100), Required Slider labels: Terrible, , Very good Custom alignment: RH
6	[reasons]	Wy are you here?	notes, Required
7	[demographics_complete]	Section Header: *Form Status* Complete?	dropdown 0 Incomplete 1 Unverified 2 Complete

Form of data



Type of analysis



What a survey is not?

SURVEY $\neq$

	ld Name	Field Label *Field Note*	Field Attributes (Field Type, Validation, Choices, Calculations, etc.)
	Demography Form (demographics) Enabled as survey		
		Study ID	text
		Sex	radio, Required 1 Female 2 Male 3 Prefer not to say
3	[age	Age (years)	text (integer, Min: 18, Max: 99), Required
4	[backgrou	Background	radio, Required 1 Social sciences 2 Medicine 3 Psychology 4 Epidemiol 5 Stati 6
5	[wellbeing]	How do you feel today?	in: 0, Max: 100), Required er labels: Terrible, , Very good Custom alignment: RH
6	[reasons]	Wy are you here?	notes, Required
7	[demographics_complete]	Section Header: *Form Status* Complete?	dropdown 0 Incomplete 1 Unverified 2 Complete

Qualitative? Quantitative?

Each cell in the grid might be filled with numeric or quantitative data (e.g. age, income, years of education, score on an IQ test, number of times assaulted etc) or it may be filled with much more qualitative information.

… we might collect quantitative data but this does not mean we are conducting a survey. If this is collected from only one case (as in a case study) or is collected only sporadically from cases we do not have the structured data set that permits survey analysis.

The logic of survey analysis is that variation in one variable is matched with variations in other variables. The notion of covariation is not an inherently statistical concept—it is a logical concept that has been systematically formulated by Mill (1879).

(De Vaus, 2002:6-7)

~~Qualitative? Quantitative?~~

PROV	SEX	POPGROUP	AGE	AGECAT10	HEIGHT	WEIGHT	WAIST	BMI
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GT	Male	White	34	(25,35]	1.87	90.2	101.5	25.8
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GT	Male	Asian	35	(25,35]	1.71	59.0	77.2	20.1
GT	Male	Asian	17	(14,25]	1.69	47.6	58.7	16.6
GT	Male	White	82	(75,120]	1.65	56.3	97.3	20.8
GT	Female	White	20	(14,25]	1.70	90.5	89.2	31.3

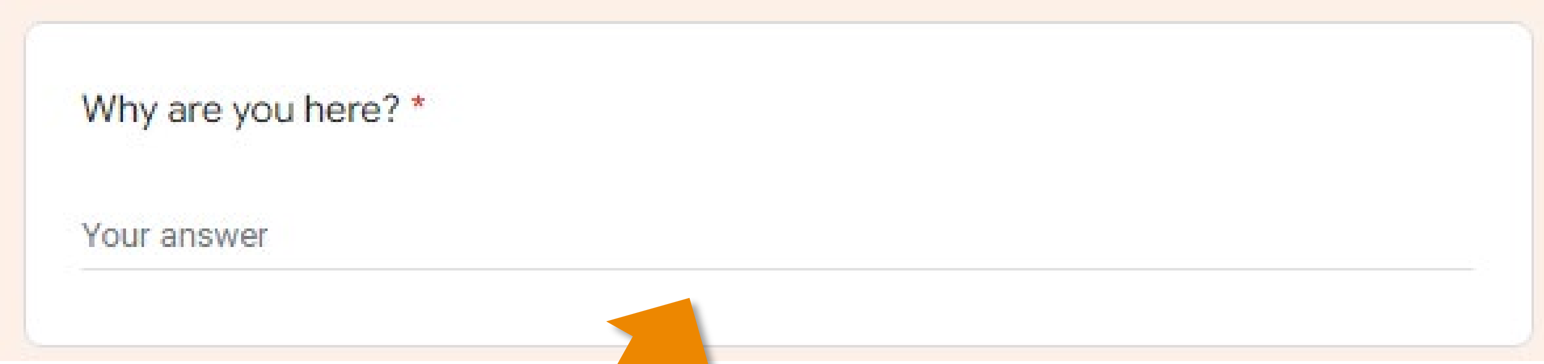
+

COMPARISION

DESCRIBE

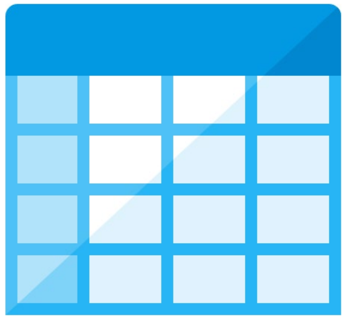
LOOK FOR RELATIONSHIPS

QUANTITATIVE DESCRIPTORS

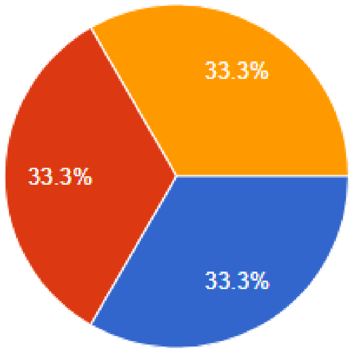

Why are you here? *

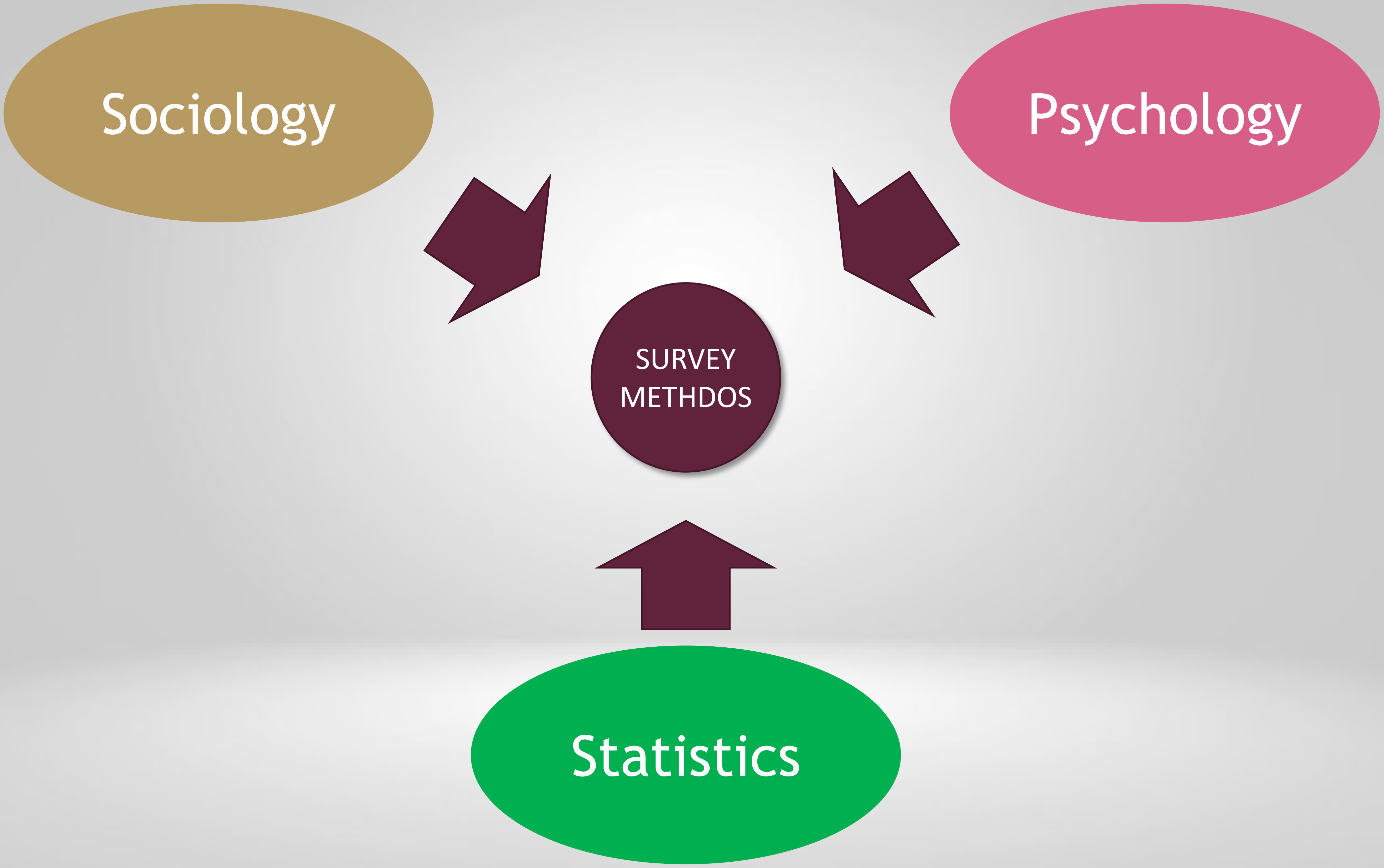
Your answer

Form of data

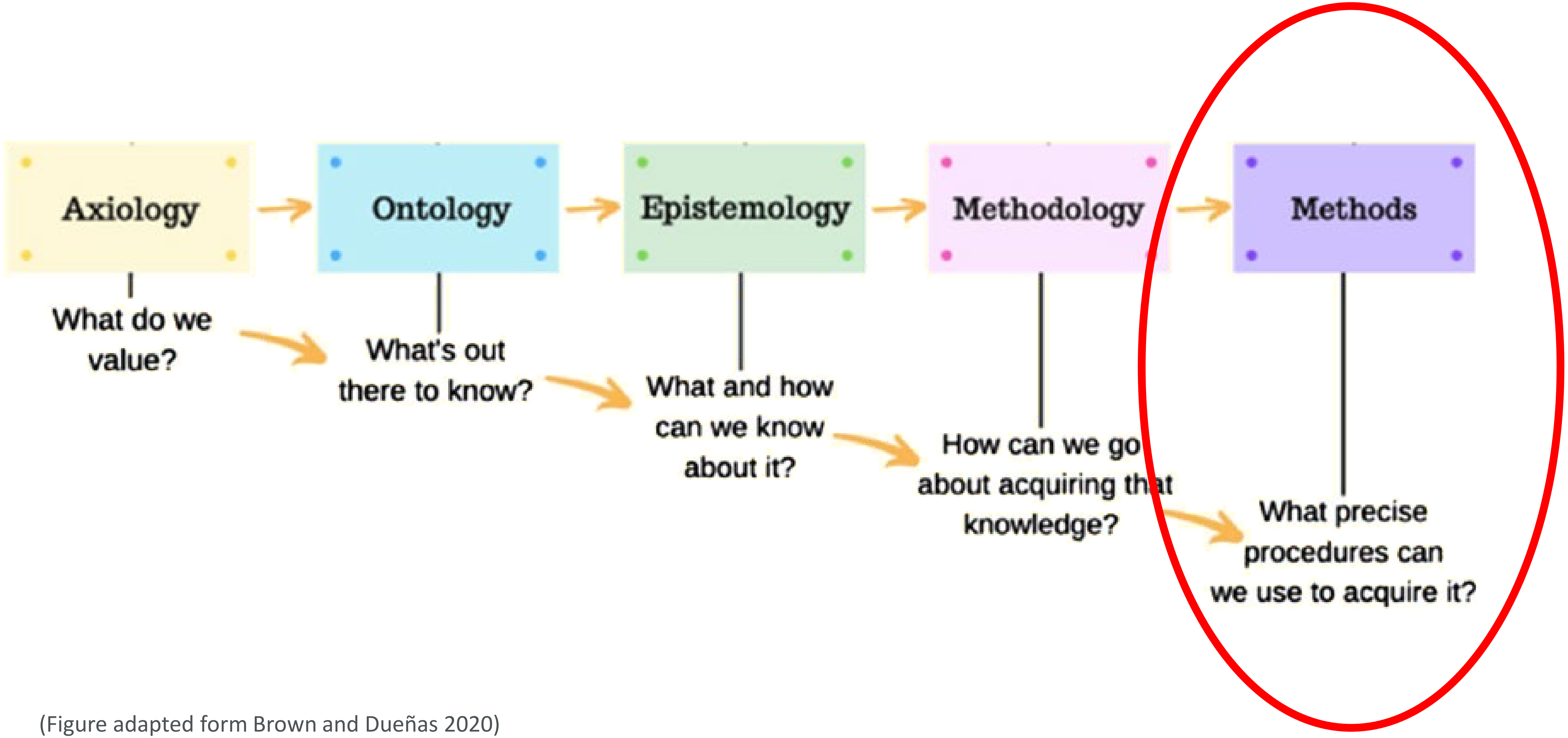


Type of analysis



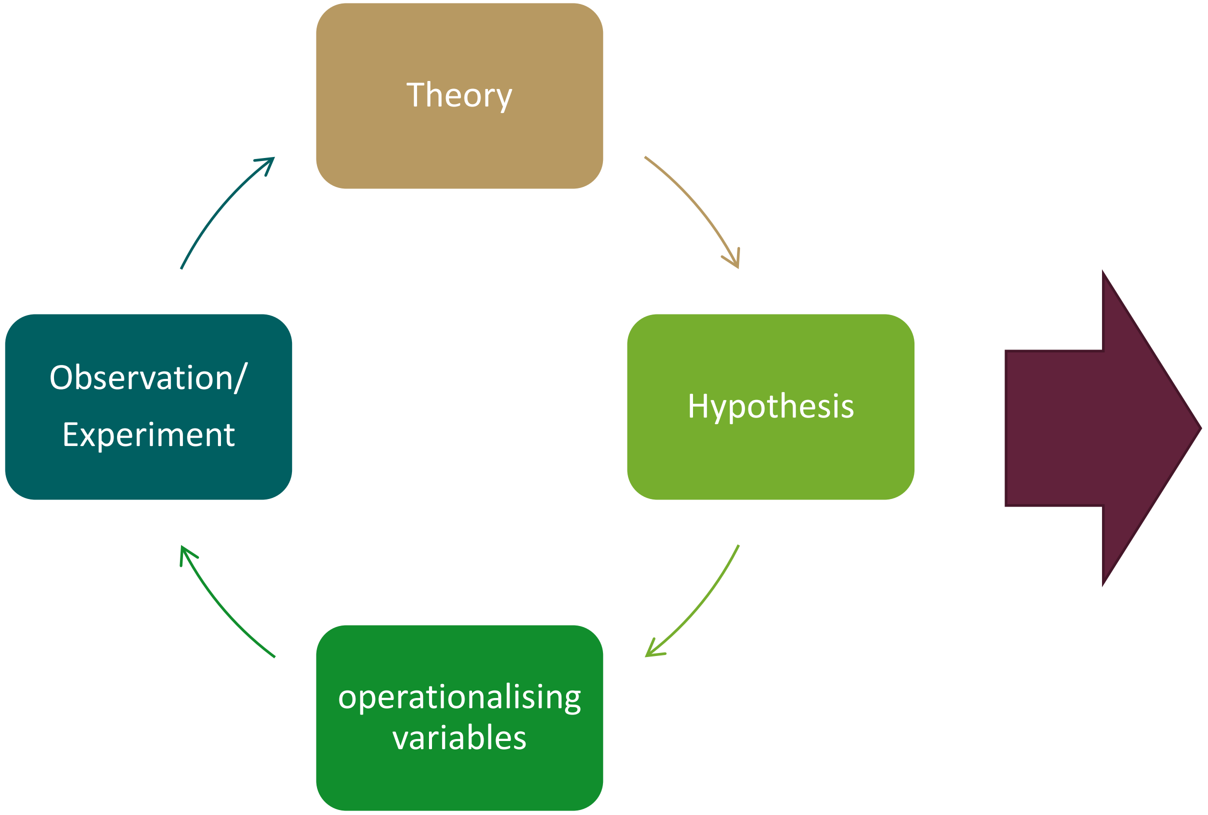

Sociology
Psychology
SURVEY METHDOS
Statistics

Surveys, theories and paradigms



(Figure adapted form Brown and Dueñas 2020)

Positivist paradigm



Positivism relies on the hypothetico-deductive method to verify a priori hypotheses that are often stated quantitatively, where functional relationships can be derived between causal and explanatory factors (independent variables) and outcomes (dependent variables)

(Park, Konge, and Artino 2020)

Positivist paradigm

Test hypotheses and control for confounding

Neutral knowledge can be obtained through the use of reliable and valid measurement tools.

There is a single, objective reality that can be observed through science.

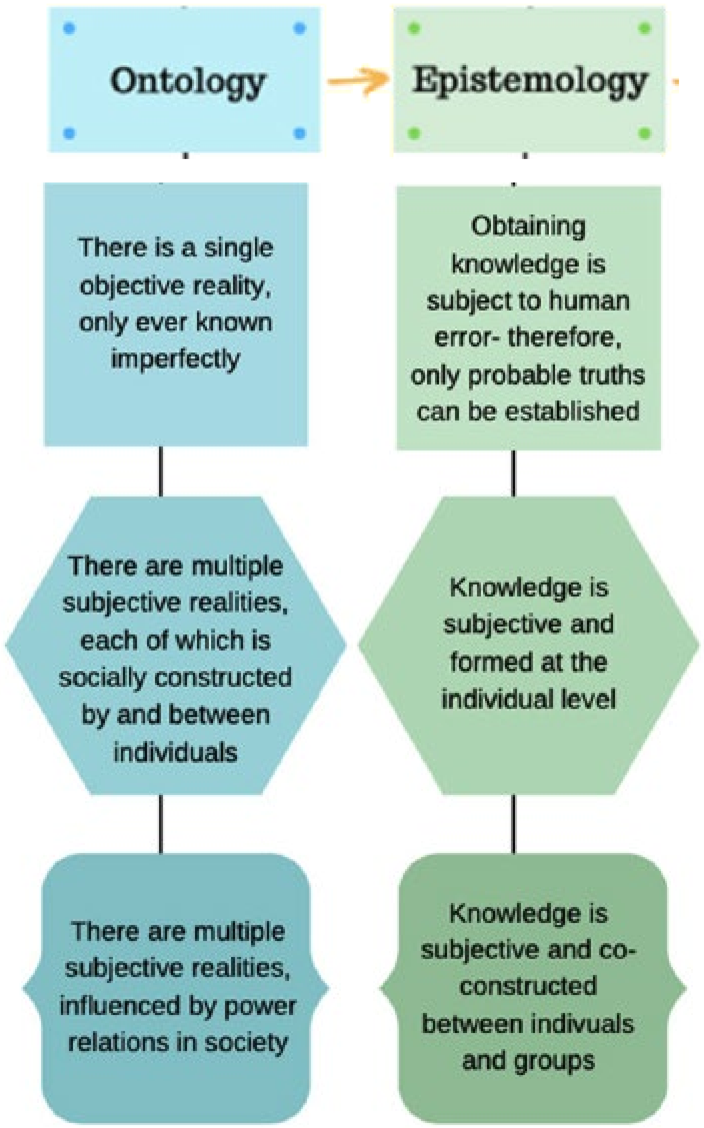
To discover laws that are generalizable and govern the universe

This is not the whole story…

POST-POSITIVISM

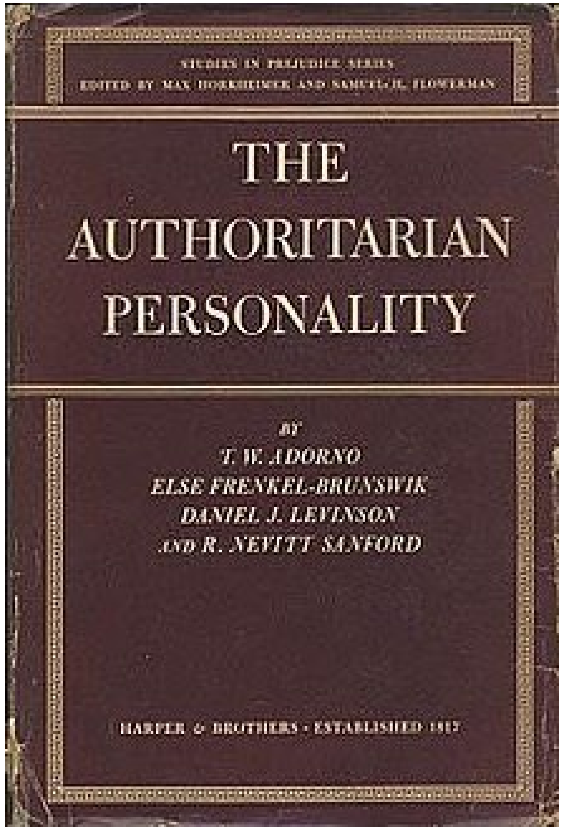
CONSTRUCTIVISM

CRITICAL THEORY



Are surveys used within these paradigms?

(Figure adapted form Brown and Dueñas 2020)



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Phenomenological Research Using a Staged Multi-Design Methodology

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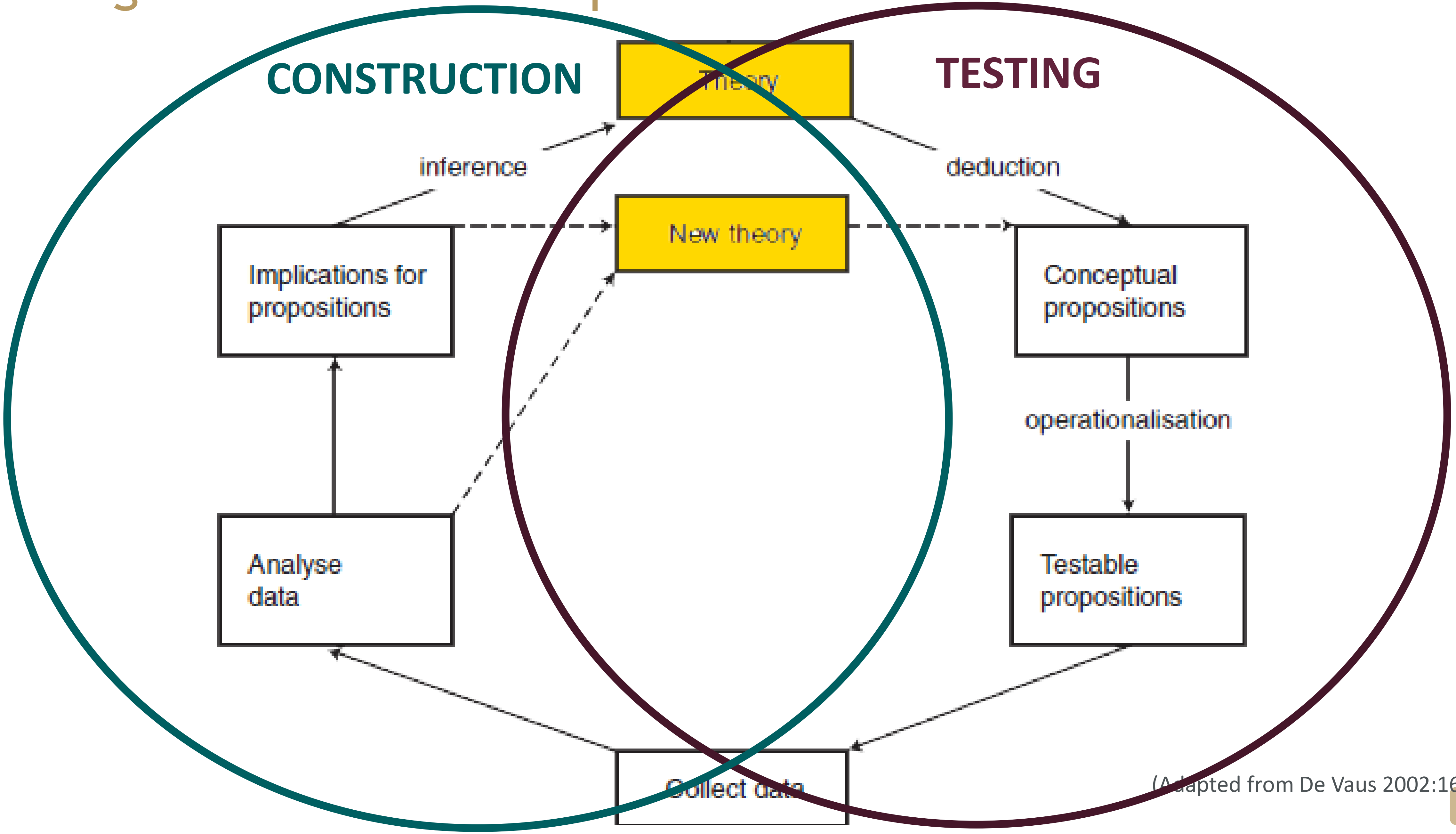
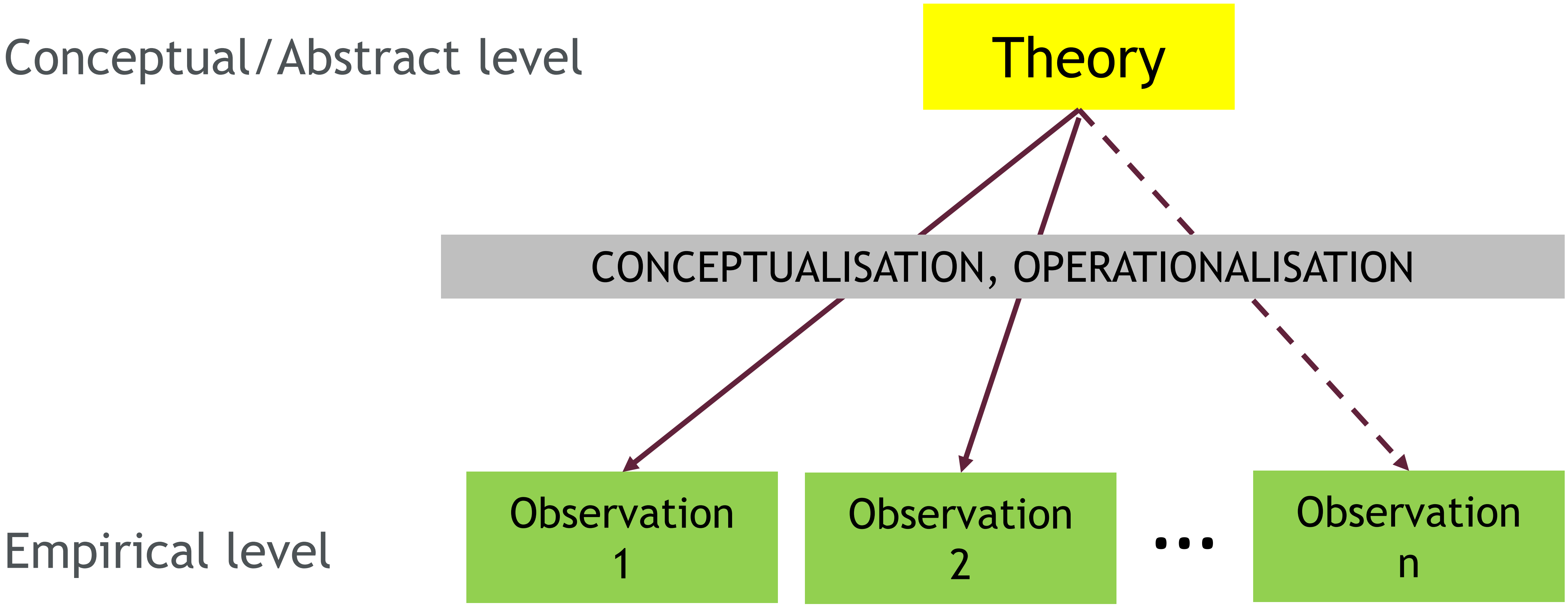
Abstract

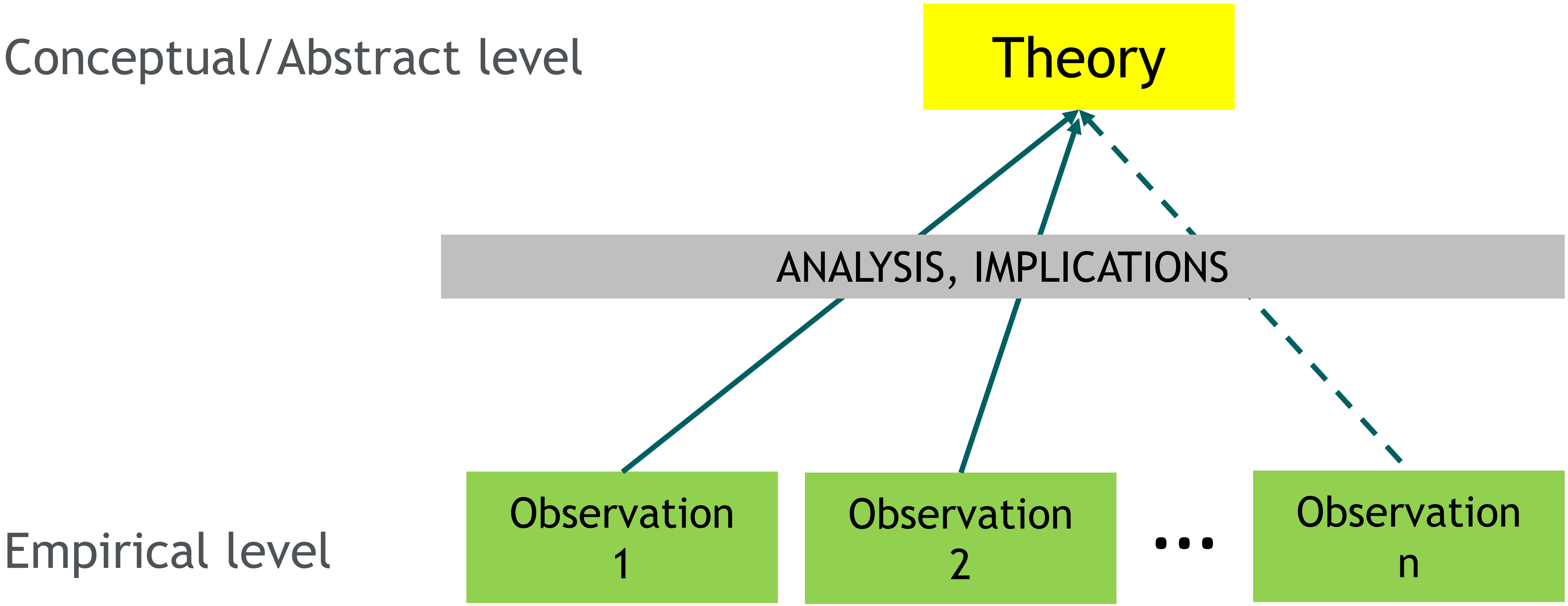
This article emphasizes the value of qualitative research using phenomenological research with a staged multi-design to provide quantitative insight, creating a mixed method research design. The original study this article is based on investigated the effects of recession in the community bank environment. The literature review includes a discussion of the banking environment to provide readers an understanding of the background for the article and for the specific methodological design.

The logic of the research process



(Adapted from De Vaus 2002:16)


Conceptual/Abstract level
Theory
CONCEPTUALISATION, OPERATIONALISATION
Empirical level
Observation 1
Observation 2
…
Observation n


Conceptual/Abstract level
Theory
ANALYSIS, IMPLICATIONS
Empirical level
Observation 1
Observation 2
...
Observation n

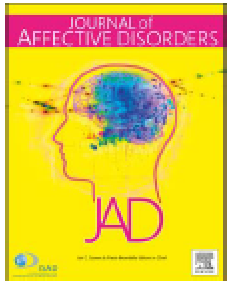
An example

Journal of Affective Disorders 229 (2018) 396–402

Contents lists available at ScienceDirect

Journal of Affective Disorders

journal homepage: www.elsevier.com/locate/jad



Research paper

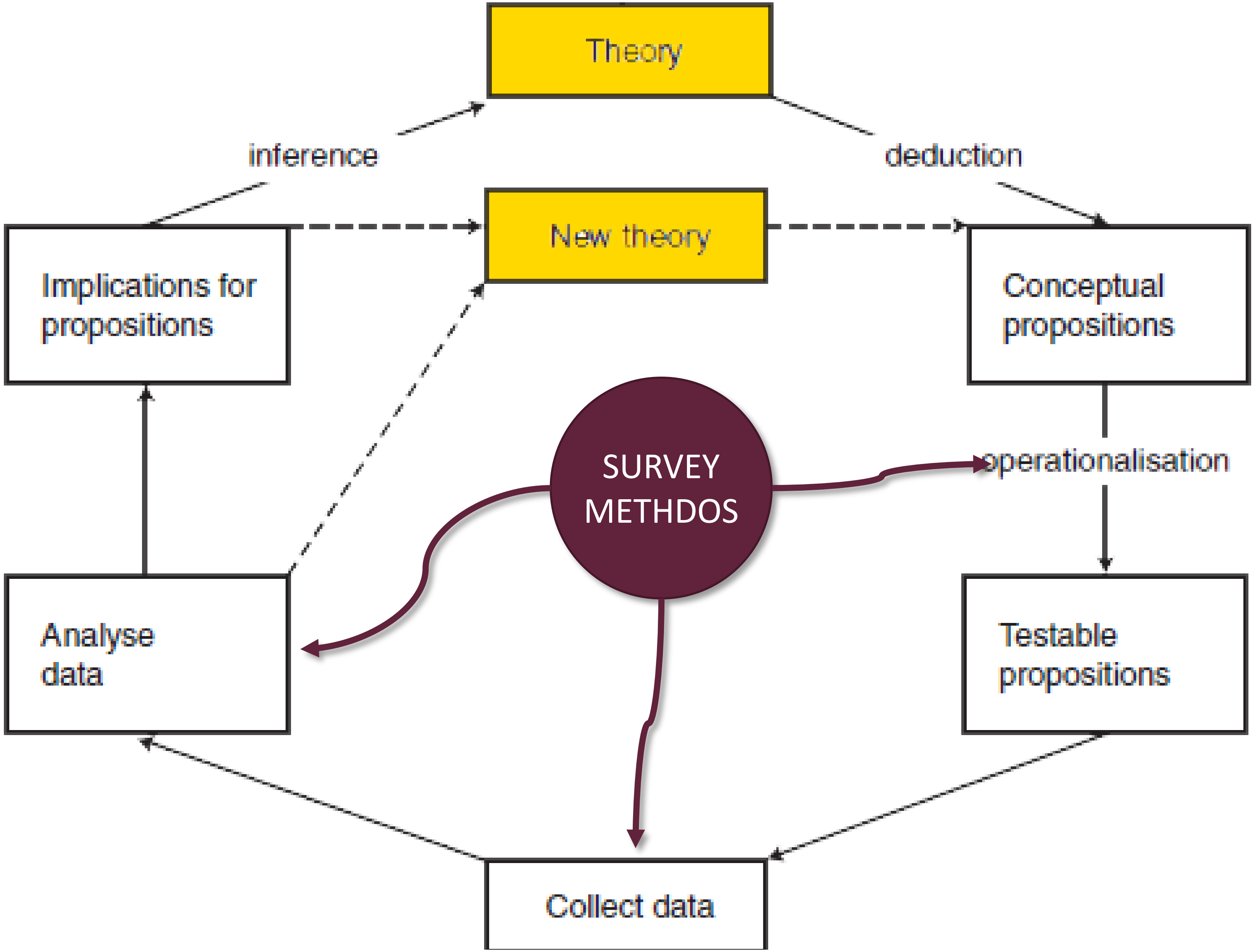
Simultaneous social causation and social drift: Longitudinal analysis of depression and poverty in South Africa

Crick Lund[a,b,*], Annibale Cois[c]

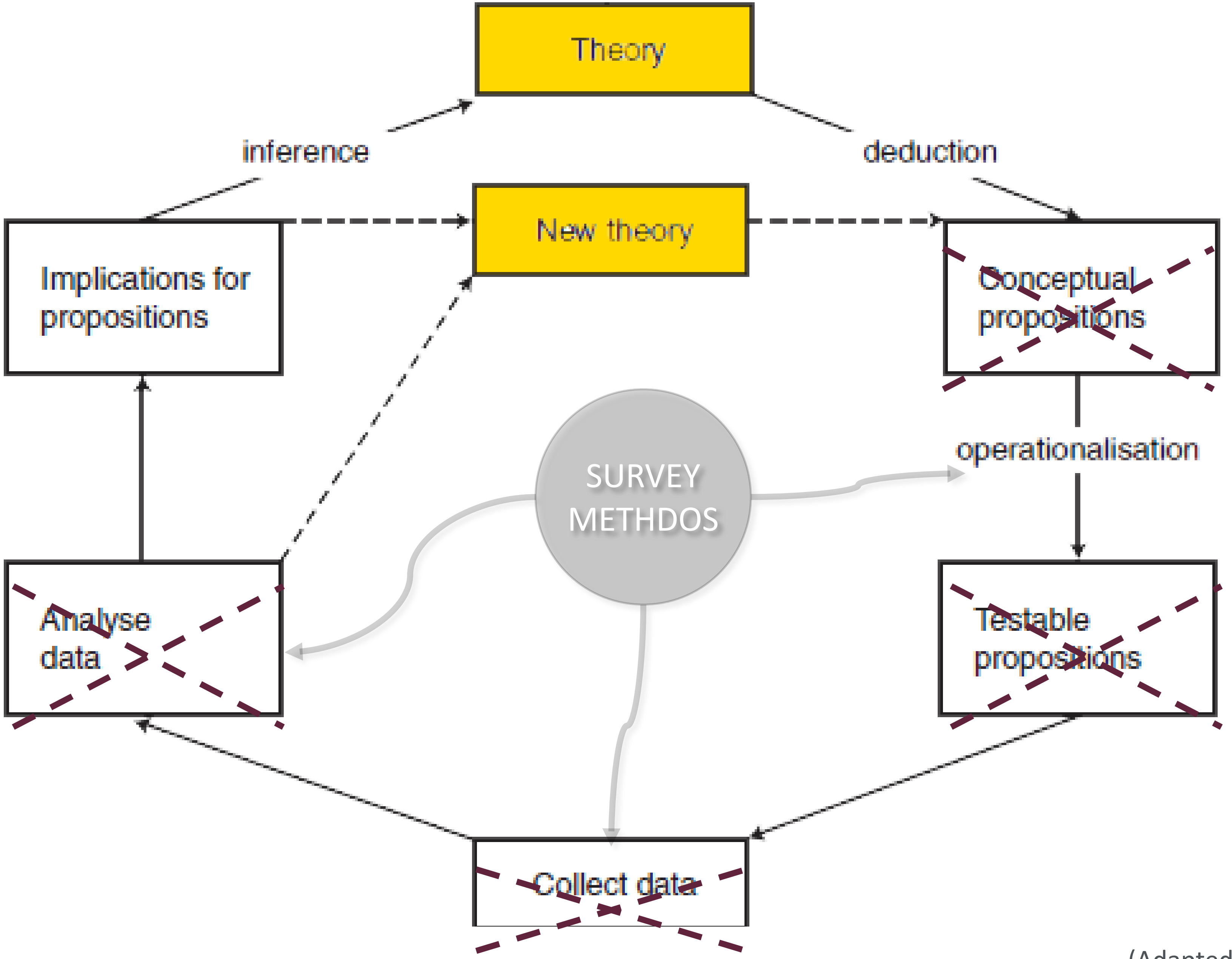
[a] Alan J Flisher Centre for Public Mental Health, Department of Psychiatry and Mental Health, University of Cape Town, South Africa
[b] Centre for Global Mental Health, Institute of Psychiatry, Psychology and Neuroscience, King's College London, United Kingdom
[c] Division of Epidemiology and Biostatistics, School of Public Health and Family Medicine, University of Cape Town, South Africa

Theory	Depression →	Poverty
Specific Proposition	The risk of future poverty is higher among people who are more depressed	
Operationalisation	Depression score	Asset index
Data collection, Analysis	Questionnaire	SEM model

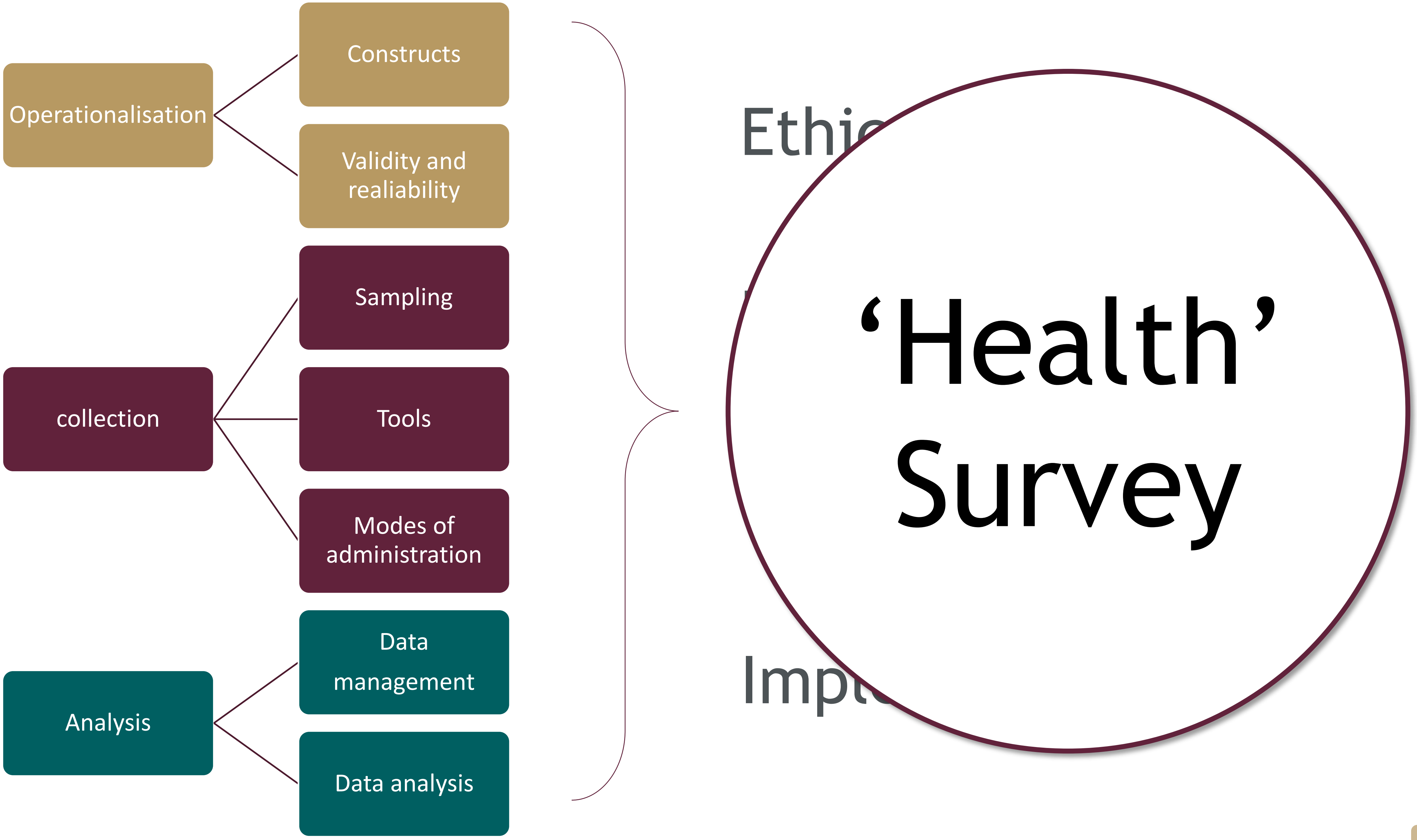
The scope of survey research



(Adapted from De Vaus 2002:16)



(Adapted from De Vaus 2002:16)


Operationalisation
Constructs
Validity and realiability
collection
Sampling
Tools
Modes of administration
Analysis
Data management
Data analysis
'Health' Survey

Summary questions

- What is a survey?
- What differentiate a survey from other research methods?
- Which kind of data collection tools do surveys use?
- What is the role of theory in the survey process?

References

De Vaus, David. 2002. *Surveys in Social Research*. Allen & Unwin. Chapter 1 & 2.

Groves, Robert M. 2011. ‘Three Eras of Survey Research’. *Public Opinion Quarterly* 75 (5): 861–71. https://doi.org/10.1093/poq/nfr057.

Park, Yoon Soo, Lars Konge, and Anthony R. Jr Artino. 2020. ‘The Positivism Paradigm of Research’. *Academic Medicine* 95 (5): 690–94. https://doi.org/10.1097/ACM.0000000000003093.

Other resources

Babbie, Earl R. 2020. *The Practice of Social Research*. Cengage learning. Chapter 2

Brown, Megan E.L., and Angelique N. Dueñas. 2019. ‘A Medical Science Educator’s Guide to Selecting a Research Paradigm: Building a Basis for Better Research’. *Medical Science Educator* 30 (1): 545–53. https://doi.org/10.1007/s40670-019-00898-9.

Lund, Crick, and Annibale Cois. 2018. ‘Simultaneous Social Causation and Social Drift: Longitudinal Analysis of Depression and Poverty in South Africa’. *Journal of Affective Disorders* 229 (January): 396–402. https://doi.org/10.1016/j.jad.2017.12.050.

Thank you

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